AlHudah Quran Platform

Complete Implementation Guide

April 2026

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Project Overview

What is AlHudah?

Mission

Provide instant access to the complete Quran with translations, tafsir, and AI-powered semantic search for authentic Islamic learning.

Vision

Build the most trusted, fast, and accessible Quran platform for Muslims worldwide — grounded in scholarship, powered by AI.

Key Features

- 6,236 verses (114 surahs)
- 30+ translations (Arabic, English, Urdu, etc.)
- Multi-scholar tafsir
- Juz-based navigation
- Bookmarking & sharing
- Dark/Light theme
- Voice search (Web Speech API)
- Responsive mobile + web

Tech Stack (Timeless Hadith Proven)

Layer	Technology	Why
Frontend	HTML + CSS + JS	Zero build, fast
Database	Supabase PostgreSQL	Free tier, REST API
Quran Data	quran.com API v4	50+ translations
AI	OpenAI GPT-4o-mini	Embeddings + Chat
Hosting	Cloudflare Pages	Free, global CDN
Auth	Supabase Auth	Email + OAuth

Infrastructure Setup

Domain & DNS Setup

1. Domain Registration

- Purchase alhudah.com
- Registrar: Namecheap or GoDaddy
- Annual cost: ~$10-15
- Enable auto-renew

2. Cloudflare Setup

- Add domain to Cloudflare
- Change NS records at registrar
- Update nameservers to Cloudflare
- Enable HTTPS + SSL/TLS
- Set Page Rules for caching

Cloudflare Pages Configuration

- 1. Connect GitHub repository to Cloudflare Pages
- 2. Select main branch for auto-deploy
- 3. Set build command: npm run build (or none for static)
- 4. Output directory: root or public/
- 5. Configure environment variables (OPENAI_API_KEY, etc.)
- 6. Enable preview deployments
- 7. Set custom domain: alhudah.com
- 8. Test deployment with git push

GitHub Repository Structure

```
alhudah/
├── index.html (home, featured verses)
├── surah.html (verse reader + translations)
├── search.html (AI-powered search)
├── about.html (mission & team)
├── bookmarks.html (user bookmarks)
│
├── css/
│   ├── main.css (base styles)
│   └── theme.css (light/dark toggle)
│
├── js/
│   ├── app.js (main app logic)
│   ├── quran-data.js (fetch from Supabase)
│   ├── tafsir.js (commentary system)
│   ├── search.js (AI search with hybrid)
│   ├── theme.js (light/dark persistence)
│   └── service-worker.js (PWA cache)
│
├── assets/
│   ├── logo.svg
│   ├── icons/
│   └── images/
│
├── api/
│   └── search-proxy.js (Cloudflare Worker)
│
├── _headers (CSP + caching rules)
├── package.json
```

Database Design

Supabase Project Setup

Create Supabase Project

- Project: alhudah-production
- Region: eu-west-1 (Europe)
- Database: PostgreSQL 14+
- pgvector extension (for embeddings)

Configure Access

- Enable Row Level Security (RLS)
- Create anon API key (read-only)
- Create service role key (admin)
- Set JWT expiry: 1 hour

Security

- Enable HTTPS only
- Restrict IP access
- Backup schedule: daily
- Monitor usage quotas

Extensions

- pg_trgm (text search)
- pgvector (AI embeddings)
- pgsodium (encryption)
- http (API calls)

Database Schema Overview

7 Core Tables + 3 Indexes

Tables:

- surahs (114 rows) — name, name_arabic, revelation_order
- ayahs (6,236 rows) — surah_id, verse_num, text_arabic, translations
- translations (30+ languages) — ayah_id, translation_name, text
- tafsir (6,236 entries) — ayah_id, scholar, commentary_text
- embeddings (6,236 vectors) — ayah_id, embedding (1536 dims)
- bookmarks (user) — user_id, ayah_id, created_at
- sessions (search history) — user_id, query, results, timestamp

Indexes:

- tsvector on ayahs.text_arabic (full-text search)
- HNSW on embeddings.embedding (vector similarity, pgvector)
- Index on bookmarks(user_id, created_at)

Key Tables: Surahs & Ayahs

Surahs (114 chapters)

Column	Type
id	INT PK
name	TEXT
name_arabic	TEXT
revelation_order	INT
verse_count	INT

Ayahs (6,236 verses)

Column	Type
id	INT PK
surah_id	INT FK
verse_num	INT
text_arabic	TEXT
juz	INT
page	INT

Data Volume

114
Surahs

6,236
Verses

30+
Languages

5
Tafsir

Row Level Security (RLS)

Public Read Access

- Surahs table: SELECT (public)
- Ayahs table: SELECT (public)
- Translations: SELECT (public)
- Tafsir: SELECT (public)
- Embeddings: NOT exposed

Authenticated Only

- Bookmarks: user_id = auth.uid()
- Sessions: user_id = auth.uid()
- Preferences: user_id = auth.uid()
- Only own records visible

Example RLS Policy

```
-- Allow public read
CREATE POLICY "Public can read"
 ON ayahs FOR SELECT
 USING (true);

-- Bookmarks only for owner
CREATE POLICY "Owner only"
 ON bookmarks FOR SELECT
 USING (user_id = auth.uid());

CREATE POLICY "Owner insert"
 ON bookmarks FOR INSERT
 WITH CHECK (user_id = auth.uid());
```

Full-Text & Vector Search Setup

Full-Text Search (FTS)

- tsvector on ayahs.text_arabic
- tsquery for search parsing
- Ranking by relevance
- Fast substring matching

Vector Search (pgvector)

- embeddings table (HNSW index)
- 1536-dim OpenAI vectors
- Semantic similarity search
- Cosine distance metric

Implementation

- CREATE INDEX idx_text_tsvector
- CREATE INDEX idx_embed_hnsw
- Test search: ~0.2s latency
- Cache results in Redis

Query Example

- SELECT * FROM ayahs WHERE
- text_arabic @@ query;
- ORDER BY ts_rank(...) DESC
- LIMIT 10;

Data Population

Quran Data Sources

Primary: quran.com API v4

- 50+ translations
- Tafsir by Ibn Kathir + others
- Word-by-word info
- Rate limit: 1000 req/day

Backup: tanzil.net

- Simple XML format
- 114 surahs, 6,236 verses
- Arabic only, highly reliable
- Offline-friendly

Tafsir Sources

- Ibn Kathir (Al-Bidayah)
- Jalalayn (summary)
- Maududi (context)
- Modern translations

Storage Format

- Supabase PostgreSQL
- UTF-8 with diacritics
- Indexed for search
- Cached at CDN level

Data Ingestion Pipeline

```
Step 1: Extract from quran.com API
  → Fetch all surahs, ayahs, translations
  → Store raw JSON locally

Step 2: Transform & Normalize
  → Map translation IDs to language codes
  → Remove duplicates, fix encoding
  → Validate verse numbers (1-6236)

Step 3: Embed with OpenAI
  → Use gpt-3.5-turbo for embeddings
  → Store 1536-dim vectors in pgvector
  → Batch: ~2000 verses per API call

Step 4: Insert into Supabase
  → Bulk insert with COPY command
  → Verify data integrity (COUNT = 6236)
  → Create indexes (tsvector, HNSW)

Step 5: Verify & Cache
  → Test search queries on live data
  → Warm Cloudflare cache
  → Monitor query performance
```

Data Verification Checklist

- `SELECT COUNT(*) FROM surahs; -- Should be 114`
- `SELECT COUNT(*) FROM ayahs; -- Should be 6,236`
- `SELECT COUNT(DISTINCT translation_id) FROM translations; -- 30+`
- `SELECT COUNT(*) FROM embeddings; -- Should be 6,236`
- `SELECT language FROM translations GROUP BY language; -- Verify all langs`
- `SELECT * FROM ayahs WHERE text_arabic IS NULL; -- No NULLs`
- `SELECT * FROM tafsir LIMIT 1; -- Check schema`
- `SELECT * FROM embeddings LIMIT 1; -- Vector format OK`
- `ANALYZE; -- Update statistics`
- `EXPLAIN ANALYZE SELECT * FROM ayahs WHERE text_arabic @@ to_tsquery('arabic:*');`

Frontend Development

Design System: Colors & Typography

Color Palette

Primary Teal — #0a6e5c

Light Teal — #1d9e8f

Gold Accent — #c8a84e

Dark BG — #060c1a

Light Gray — #f5f5f7

Typography

Element	Font	Size
Page Title	Cairo Play	44pt bold
Section Header	Cairo Play	28pt bold
Body Text	Droid Sans	16pt
Verse Text (AR)	Amiri	18pt
Translation (EN)	Droid Sans	14pt
Caption	Droid Sans	12pt muted

Core Page Layouts

Landing Page

- Hero: featured verse + CTA
- Recent surahs grid
- Search bar (prominent)
- Dark theme default

Surah Reader

- Sidebar: surah list
- Main: verses + translations
- Tafsir panel (toggle)
- Bookmark button each verse

Search Page

- Search bar (full width)
- Results list (cards)
- Filters: language, surah
- AI chat mode toggle

Bookmarks Page

- Grid or list view
- Sort: date, surah, rating
- Export to PDF feature
- Share collection link

Landing Page Structure

```
┌──────────────────────────────────────┐
│ Header: Logo + Nav + Theme Toggle    │
├──────────────────────────────────────┤
│                                      │
│ Hero Section:                        │
│ "Explore the Quran"                  │
│ Featured Verse + Translation         │
│ Search Bar (prominent)               │
│                                      │
├──────────────────────────────────────┤
│ Recent Surahs Grid (6 items)         │
│ Al-Fatihah, Al-Baqarah, etc.         │
├──────────────────────────────────────┤
│ Stats: 114 Surahs, 6,236 Verses      │
│ 30+ Translations, 5 Tafsir           │
├──────────────────────────────────────┤
│ CTA: "Start Reading Now"             │
├──────────────────────────────────────┤
│ Footer: Links, Social, Contact       │
└──────────────────────────────────────┘
```

Surah Reader Page Layout

```
┌──────────────────────────────────────────┐
│ Header: Surah Name + Juz Nav             │
├──────────┬───────────────────────────────┤
│          │ Verse 1:1 (Arabic)            │
│ Sidebar: │ [Translation English]         │
│ Surah    │ Tafsir (toggle)               │
│ List     │ [Bookmark] [Share]            │
│          │                               │
│          │ Verse 1:2 (Arabic)            │
│ - Al-    │ [Translation English]         │
│  Fatih   │ ...                           │
│ - Al-    │                               │
│  Baqar   │ Verse 1:286 (last)            │
│ - ...    │ [Bookmark] [Share]            │
│          │                               │
│ [Search] │ ← pagination/scroll →         │
└──────────┴───────────────────────────────┘
```

Light / Dark Theme System

Theme Architecture

- CSS custom properties (vars)
- data-theme attribute
- localStorage persistence
- No theme flicker on load

CSS Variables

- --color-bg-primary
- --color-text-primary
- --color-accent-teal
- --color-border

Light Mode

- Background: #ffffff
- Text: #1a1a1a
- Accent: #0a6e5c (teal)
- Borders: #e0e0e0

Dark Mode

- Background: #0f0f0f
- Text: #f5f5f7
- Accent: #1d9e8f (light teal)
- Borders: #333333

PWA & Service Worker

Service Worker Features

- Cache static assets
- Offline fallback
- Background sync
- Push notifications

Manifest.json

- Name: AlHudah Quran
- Icon: 192x192, 512x512
- Theme color: #0a6e5c
- Display: standalone

Caching Strategy

- Verses: Cache-first (1 week)
- Translations: Stale-while-revalidate
- User data: Network-first
- API responses: 5-minute cache

Install Prompt

- Show after 2nd visit
- Bottom banner (subtle)
- 'Install to Home Screen'
- Dismissible, no forced UX

Quran Reader Features

Surah List Display

Grid Layout (2-3 columns on mobile, 3-4 on desktop)

```
┌──────────────────┐ ┌──────────────────┐
│ 1. Al-Fatihah    │ │ 2. Al-Baqarah    │
│ The Opening      │ │ The Heifer       │
│ 7 verses         │ │ 286 verses       │
│ Meccan           │ │ Madinan          │
│ 🔖 Bookmark      │ │ 🔖 Bookmark      │
└──────────────────┘ └──────────────────┘

┌──────────────────┐ ┌──────────────────┐
│ 3. Al-'Imran     │ │ 4. An-Nisa'      │
│ The Family...    │ │ The Women        │
│ 200 verses       │ │ 176 verses       │
│ Madinan          │ │ Madinan          │
│ 🔖 Bookmark      │ │ 🔖 Bookmark      │
└──────────────────┘ └──────────────────┘
```

Features:

- Search / filter by name
- Sort: revelation order, alphabetical
- Bookmark count badge per surah
- Arabic names + English translations
- Responsive grid (1 col mobile)

Verse Display Card Layout

```
┌──────────────────────────────────────┐
│ Surah 1, Verse 1 (Juz 1, Page 1)     │ ← Metadata
├──────────────────────────────────────┤
│ الحمد لله رب العالمين                │ ← Arabic (Amiri, 20pt)
├──────────────────────────────────────┤
│ All praise is due to Allah, the      │ ← Translation (Droid, 14pt)
│ Lord of all the worlds.              │
│ — Sahih International                │ ← Source attribution
├──────────────────────────────────────┤
│ ٱلْحَمْدُ لِلَّهِ رَبِّ ٱلْعَٰلَمِينَ │ ← Word-by-word (12pt)
│ All-praise (be) to Allah             │
│ Lord of the worlds                   │
├──────────────────────────────────────┤
│ [📖 Tafsir] [🔖 Bookmark] [Share]    │ ← Actions
└──────────────────────────────────────┘
```

Interactions:

- Click verse → expand tafsir panel
- Click bookmark icon → toggle bookmarked state
- Click share → copy link or social share
- Swipe (mobile) → next verse
- Long press → select text for copy

Translation Toggles & Juz Navigation

Translation Switching

- Dropdown: 30+ languages
- Default: Sahih International
- Urdu, Arabic, French, etc.
- Instantly load from Supabase

Language Support

- English (3 major translations)
- Urdu, Bengali, Tajik
- Spanish, French, German
- Bosnian, Albanian, Turkish

Juz Navigation

- Juz selector (30 options)
- Quick jump to Juz 1-30
- Shows verse range per Juz
- Keyboard shortcut: J

Bookmarking

- Bookmark count badge
- Persistent in Supabase
- Edit collections (optional)
- Export bookmarks as JSON

Mobile Interactions & Responsive

Touch Gestures

- Swipe left → next verse
- Swipe right → previous verse
- Pinch → zoom verse text
- Long press → copy text

Mobile Layout

- Single column (verses full-width)
- Stacked: Arabic, Translation
- Actions below verse
- Drawer menu for nav

Performance

- Lazy load verses on scroll
- Virtual scroll (for large lists)
- Min 60fps animations
- Optimized font loading

Accessibility

- Touch targets ≥48px
- High contrast text (WCAG AA)
- Screen reader support
- Keyboard navigation (Tab/Arrows)

Tafsir System

Supported Tafsir Scholars

Scholar	Work	Scope
Ibn Kathir	Al-Bidayah wa'n-Nihayah	Full (6,236 verses)
Jalalayn	Al-Jalalayn (Brief)	All surahs
Maududi	Tafhim ul-Quran	Thematic interpretation
At-Tabari	Tafsir At-Tabari	Classical (partial)
An-Nawawi	Al-Manzil	Jurisprudence-focused

Tafsir API Integration & Caching

```
Three-Level Caching Strategy:

Level 1: Browser Cache (localStorage)
  └─ Store last 50 tafsir loads (5 MB)
  └─ Expires after 7 days
  └─ Instant access (0ms)

Level 2: Supabase Cache (DB)
  └─ All 6,236 tafsir entries stored
  └─ Indexed by ayah_id
  └─ ~200ms query time

Level 3: quran.com API (on-demand)
  └─ Fallback for missing tafsir
  └─ Rate-limited: 1000 req/day
  └─ Cache result in Supabase

Implementation:
  1. User clicks "Tafsir" on verse 2:255
  2. Check localStorage for tafsir:255
  3. If miss → Query Supabase
  4. If miss → Call quran.com API + save
  5. Display with scholar attribution
  6. Cache in browser for offline access
```

AI-Powered Search

Hybrid Search: Semantic + Full-Text

```
User Query: "What does the Quran say about kindness to parents?"

│
├─ Path 1: Full-Text Search (Supabase tsvector)
│  └─ SELECT ayahs WHERE text_arabic @@ to_tsquery('parents|kindness')
│  └─ Fast, keyword-based, high recall
│  └─ Returns: Ayah 17:23 (Surah Al-Isra)
│
└─ Path 2: Vector Search (OpenAI Embeddings)
   └─ Embed query with OpenAI API
   └─ Query pgvector for similar verses
   └─ Cosine similarity threshold: 0.75
   └─ Returns: Ayah 46:15 (Surah Al-Ahqaf)

│
└─ Merge Results (Hybrid Ranking)
   ├─ Full-text score: 0.92 (keyword match)
   ├─ Vector score: 0.88 (semantic match)
   ├─ Combined: 0.90 (ranked #1)
   │
   ├─ Return top 10 merged results
   ├─ Display with highlighting
   └─ Show context + translation
```

Cloudflare Worker API Proxy

Purpose

- Protect OpenAI API key
- Rate limit requests (10/min/user)
- Cache embedding responses
- Log queries for analytics

Execution

- Runs at edge (Cloudflare CDN)
- Zero cold start latency
- Automatic scaling
- Free tier: 100,000 req/month

Request Flow

- Browser → POST /api/search
- Worker checks API key
- Worker calls OpenAI
- Worker queries Supabase
- Return merged results

Security

- API key never exposed
- JWT validation
- CORS headers set
- DDoS protection enabled

OpenAI API Integration

```
Model: gpt-3.5-turbo
  └─ Embeddings: text-embedding-3-small
    ├─ Dimensions: 1536
    ├─ Cost: $0.02 per 1M tokens
    └─ Speed: ~50ms per request

ChatCompletions: gpt-4o-mini
  └─ For natural language Q&A
  └─ System prompt: Islamic scholar tone
  └─ Temperature: 0.7 (balanced creativity)
  └─ Max tokens: 500

Usage Limits:
  ├─ Embeddings: 6,236 vectors (one-time cost)
  ├─ Searches: ~$0.01 per search (average)
  ├─ Monthly budget: $10-50 (estimated)
  └─ Monitoring: Track usage via OpenAI dashboard

Integration Code:
  const response = await fetch('/api/search', {
    method: 'POST',
    body: JSON.stringify({ query: userQuery })
  });
  const results = await response.json();
```

Voice Search with Web Speech API

Web Speech API

- Browser-native (no API key)
- Supports 100+ languages
- Real-time transcription
- Offline capable

How It Works

- User clicks mic button
- Browser prompts permission
- Listen for 30 seconds
- Transcribe to text

Implementation

- const SpeechRecognition = ...
- recognition.start()
- recognition.onresult = ...
- Pass result to search API

User Experience

- Visual waveform feedback
- Show transcript as typing
- Cancel button (always visible)
- Language selector (Arabic/English)

SEO & Performance

Meta Tags & Structured Data

```
HTML Meta Tags (Head):
  <title>Al-Fatihah - Surah 1 | AlHudah Quran</title>
  <meta name="description" content="Read Surah Al-Fatihah (Chapter 1) of the Quran with ...">
  <meta name="keywords" content="quran, surah, al-fatihah, tafsir, translation">
  <meta property="og:title" content="Al-Fatihah | AlHudah">
  <meta property="og:image" content="/og-image-1.jpg">
  <meta name="twitter:card" content="summary_large_image">

JSON-LD Schema (Structured Data):
  {
    "@context": "https://schema.org",
    "@type": "Book",
    "name": "The Quran",
    "isbn": "978-0-00000-000-0",
    "author": { "@type": "Person", "name": "Allah" },
    "hasPart": [ { "name": "Al-Fatihah", "text": "..." } ]
  }

Sitemap.xml:
  <url>
    <loc>https://alhudah.com/surah/1</loc>
    <lastmod>2026-04-13</lastmod>
    <changefreq>monthly</changefreq>
    <priority>0.8</priority>
  </url>
```

Core Web Vitals & Lighthouse 90+

Metric	Target	Strategy
LCP (Largest Contentful Paint)	< 2.5s	Lazy load images, minimize JS
FID (First Input Delay)	< 100ms	Defer non-critical scripts
CLS (Cumulative Layout Shift)	< 0.1	Preload fonts, set image sizes
TTFB (Time to First Byte)	< 600ms	CDN caching, optimized DB
Overall Lighthouse	> 90	Remove render-blocking resources

Deployment & CI/CD

Cloudflare Pages Auto-Deploy Pipeline

```
GitHub Push (main branch)
  │
  └─ Webhook trigger
     │
     └─ Cloudflare Pages receives event
        │
        ├─ Run build command (if configured)
        │  └─ npm run build or npm run minify
        │
        ├─ Upload artifacts to Cloudflare edge
        │  └─ Global CDN distribution
        │
        ├─ Create preview deployment URL
        │  └─ https://[branch].[project].pages.dev
        │
        ├─ Activate as production
        │  └─ Available at alhudah.com (via CNAME)
        │
        └─ Cache invalidation
           └─ Instant purge of old assets

Timeframe: 30-60 seconds (end-to-end)
Rollback: Simple git revert + push (instant)
Monitoring: Real-time logs in dashboard
```

GitHub Actions Validation Pipeline

```
Workflow: .github/workflows/ci.yml

Trigger: On every push and PR to main

Jobs:
 1. Lint & Format Check
    └─ eslint on js/
    └─ stylelint on css/
    └─ html validator on html/

 2. Security Scan
    └─ npm audit (dependencies)
    └─ Check for exposed secrets
    └─ CSP header validation

 3. Lighthouse Audit
    └─ Run on https://alhudah.pages.dev
    └─ Verify Lighthouse > 85
    └─ Report Core Web Vitals

 4. Accessibility Check
    └─ axe accessibility scan
    └─ WCAG 2.1 AA validation

 5. Link Checker
    └─ Verify no broken links
    └─ Check internal anchors

Failure → Blocks merge to main
Success → Auto-deploy to Cloudflare Pages
```

Maintenance & Costs

Monthly Cost Breakdown

Service	Cost/Month	Notes
Domain (alhudah.com)	$1	Annual: ~$12
Cloudflare Pages	$0	Free (100k req/mo)
Supabase (PostgreSQL)	$0-25	Free tier: 500 MB storage
OpenAI Embeddings	$1-5	6,236 vectors (one-time) + searches
Supabase Storage (backups)	$0-5	Automatic daily backups
Monitoring & Analytics	$0	Free: Vercel, Sentry
TOTAL	$2-35	Scalable as traffic grows

Implementation Timeline (Phases 0-10)

Phase	Task	Duration
0	Repository analysis & planning	1 day
1	Infrastructure setup (GitHub, Cloudflare, Supabase)	2 days
2	Database schema & data population (6,236 verses)	3 days
3	Frontend foundation (HTML, CSS, JS)	3 days
4	Quran reader (surahs, verses, translations)	4 days
5	Tafsir system (multi-scholar, caching)	2 days
6	AI search (embeddings, hybrid search, chat)	3 days
7	SEO & performance optimization	2 days
8	Deployment & CI/CD pipeline setup	2 days
9	Final QA & polish	2 days
10	Final report & documentation	1 day

Total: ~25 days (assuming 8-hour days)

Key Lessons from Timeless Hadith

Technical Wins

- Supabase REST API is rock-solid
- Cloudflare Pages auto-deploy works great
- Full-text search with tsvector is fast
- Vanilla JS (no framework) is plenty

Content Strategy

- Source data from reliable APIs
- Never invent or hallucinate
- Cache aggressively
- Cite sources always

Design Learnings

- Dark theme default popular with users
- Theme persistence critical (localStorage)
- Remove emojis for premium feel
- Responsive mobile-first design essential

Pitfalls to Avoid

- CSP headers can block fonts (fix early)
- Always preload critical resources
- Test theme on slow networks
- Monitor API rate limits closely

Thank You

Ready to Launch AlHudah?

https://alhudah.com

GitHub: ehsaasradio-bot/alhudah

Contact: ehsaasradio@gmail.com